

JASON LAI, PH. D.

Data Scientist (Senior Manager)

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Work Location: Singapore/Australia

SUMMARY

Senior data scientist with 5 years of industry experience and 7 years of research experience. I have made statistical models for lung cancer biomarker identification used in on-going clinical trials, as well as end-to-end delivery of 2 credit risk models for the consumer card business in Europe and ANZ. Versatile problem-solver, collaborative and curiosity-driven.

SKILLS

Coding Languages	Python, R, SQL, BigQuery
ML/AI Toolbox	Predictive modeling (xgboost, sklearn, tensorflow); Explainability (shap, eli5); Hyperparameter tuning (optuna, bayesopt, gridsearch); Data wrangling (pandas, pyspark, tidy); Visualization (seaborn, ggplot, shiny)
HPC Stack	Workload managers (slurm, SGE); Containers (docker, singularity); Pipeline (nextflow)
Cloud Computing	AWS, GCP
Statistics	Regression, ANOVA, PCA, Clustering, Pearson/Spearman Correlation
Communication	English (Native), Mandarin, German & Malay

EXPERIENCE

American Express

Senior Manager, Data Science

Manager, Data Science

Singapore

Jun 2024 - Present

May 2023 - May 2024

Responsibilities

- Lead end-to-end development and deployment of credit risk models for prospect underwriting of the international consumer card business.
- Engage with credit risk managers to address opportunity and business focus areas by engineering features to capture risk behaviors of vulnerable segments.
- Coordinate with governance, compliance and privacy partners to manage model risk.

Achievements

- Incorporated survival-based framework, new credit bureau features and bank-based risk indices into credit risk models for the consumer portfolio in Europe and ANZ that improved risk discrimination by up to 25% and 2-year incremental PTI of \$7M. (Excellence Award winner)
- Delivered a proof-of-concept propensity model to drive a targeted marketing campaign that increased revolving credit by 27%.
- Developed a semi-supervised learning framework for Reject Inference that improved prospect credit risk models by 10%.

Hummingbird Bioscience

(Hummingbird Bioscience designs antibody for precision therapeutics. Her lead asset, HMBD-001, is undergoing Phase II clinical trial for lung cancer treatment.)

Machine Learning and Bioinformatics Scientist

Singapore

Aug 2021 - Apr 2023

Responsibilities

- Led the Bioinformatics team to deliver biological insights from high-throughput genetic data of laboratory and clinical experiments.
- Collaborated with medical professionals and clinical trial program managers to identify biomarkers, market sizing and support the design of patient selection strategy.

Achievements

- Delivered a purpose-built data pipeline with Nextflow implemented on-prem to enable in-house high-throughput genomics experiments.
- Identified 3 genetic mutation signatures of lung cancer that associates with response to company's proprietary therapeutic asset.

EDUCATION

Ph. D., Natural Sciences, Goethe University Frankfurt am Main 2014-2018
B. Sc. (Hons.), Life Sciences and Statistics, National University of Singapore 2008-2012
Study Abroad Program, University of California, Berkeley (Spring 2011)

RESEARCH EXPERIENCE

Mechanobiology Institute *Singapore*
Research Fellow May 2018 - Jul 2021

- Led the research of the Hippo pathway in skin development, resulting in 1 peer-reviewed publication.
- Implemented dashboards for intuitive data exploration of gene function in developing zebrafish embryos.

Max Planck Institute *Bad Nauheim, Germany*
Graduate Researcher Feb 2014 - Jan 2018

- Conducted independent research on the Hippo pathway in cardiovascular development and morphogenesis, resulting in 3 peer-reviewed articles and presentations at 3 international conferences.
- Supported bioinformatics data processing and analyses.

Saw Swee Hock School of Public Health *Singapore*
Research Assistant Jul 2012 - Jan 2014

- Assisted senior scientists in genetic variant annotation of ancestry- and disease-related information.